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## Chapter 5

# The Things Topology Are

5.1 Problem 5.1

5.2 Problem 5.2

## 5.3 Problem 5.3

For each of the following, provide a proof or a counterexample.

- (a) If  $A$  and  $B$  are compact, must  $A \cup B$  be compact?
- (b) If  $A$  and  $B$  are compact, must  $A \cap B$  be compact?

### 5.3.1 Solution (a)

*Proof.* By the Heine-Borel Theorem, if  $A$  and  $B$  are compact, the two of them are closed and bounded. We can name bounds for the two of these and recall the definition of being closed (having an open complement).

$$\forall a \in A, L_A \leq a \leq U_A \quad (5.1)$$

$$\forall b \in B, L_B \leq b \leq U_B \quad (5.2)$$

$$\forall x \in A^c, \exists \delta > 0 : (x - \delta, x + \delta) \subseteq A^c \quad (5.3)$$

$$\forall x \in B^c, \exists \delta > 0 : (x - \delta, x + \delta) \subseteq B^c \quad (5.4)$$

Suppose we define the union of  $A$  and  $B$  and its complement by DeMorgan's Law.

$$A \cup B = \{c | c \in A \vee c \in B\} \quad (5.5)$$

$$(A \cup B)^c = A^c \cap B^c \quad (5.6)$$

Since both  $A^c$  and  $B^c$  are open and the intersection of a finite number of open sets is also open,  $A^c \cap B^c$  is open. By extension,  $(A \cup B)^c$  is open and  $A \cup B$  is closed.

Suppose that  $U = \max(U_A, U_B)$  and  $L = \min(L_A, L_B)$ .

$$\forall a \in A, U \geq U_A \geq a \quad (5.7)$$

$$\forall b \in B, U \geq U_B \geq b \quad (5.8)$$

$$\forall a \in A, L \leq L_A \leq a \quad (5.9)$$

$$\forall b \in B, L \leq L_B \leq b \quad (5.10)$$

From this, we conclude that  $U$  is an upper bound of  $A \cup B$  and  $L$  is a lower bound of  $A \cup B$ . This means that  $A \cup B$  is bounded. Since  $A \cup B$  is both closed and bounded, then by the Heine-Borel Theorem, it is compact.  $\square$

### 5.3.2 Solution (b)

*Proof.* By the Heine-Borel Theorem, if  $A$  and  $B$  are compact, the two of them are closed and bounded. This means that  $A^c$  and  $B^c$  are open. This means that  $A^c \cup B^c$ , which can be turned into  $(A \cap B)^c$  by DeMorgan's Laws, is open.  $A \cap B$  is closed.

We can write the definition of  $A \cap B$  and name bounds for the two of these.

$$A \cap B = \{c | c \in A \text{ and } c \in B\} \quad (5.11)$$

$$\forall a \in A, L_A \leq a \leq U_A \quad (5.12)$$

$$\forall b \in B, L_B \leq b \leq U_B \quad (5.13)$$

$$\forall c \in A \cap B, L_A \leq c \leq U_A \text{ and } L_B \leq c \leq U_B \quad (5.14)$$

Since there is a minimum and maximum bound of  $c$ , we can say that  $A \cap B$  is bounded.

Since a bounded and closed set is compact,  $A \cap B$  is compact. OEΔ

## 5.4 Problem 5.5

- (a) Give an example of countably many disjoint open intervals.
- (b) Prove that there does not exist a collection of countably many disjoint open intervals.

## 5.5 Problem 5.7